Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/28/2019
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	2012 Presidential Early Career Award for Scientist and Engineers (PECASE)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	joshua.s.alwood@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Space Biology PECASE AWARD
Start Date:	04/01/2015	**End Date:**	10/01/2018
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	16	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	End date changed to 10/1/2018 to accommodate final reporting (Ed., 3/4/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts are exposed to weightlessness and ionizing radiation during spaceflight. To combat the bone loss that results from weightlessness, astronauts spend a substantial amount of time exercising or, in the future, will be exposed to artificial gravity as a countermeasure. We aim to study these environmental conditions using basic biological models in ground-based work that lays the ground work for future work. Throughout life, bone remodeling is intimately tied to sex steroid hormones and their activated receptors. In females, estrogen receptors regulate growth plate closure, cessation of periosteal and endosteal bone growth, and cessation of cancellous bone resorption, among other things. Whether sex-steroid hormones and their receptors are modulated in skeletal tissue during long-duration spaceflight is unclear. This Presidential Early Career Award for Scientist and Engineers (PECASE) award consisted of five sub-projects: Project 1, Title: Epigenetic Regulation During Simulated Spaceflight: MicroRNA Expression and Methylation-Specific Sequencing. Description and Aim: This work complements the methylation work performed in the previous Space Biology New Investigator Grant ("Bone Loss During Simulated Weightlessness: The Role of Osteoclasts", ended 2015), by allowing ‘omics approaches to cover larger swaths of the genome (e.g., whole genome sequencing and RNA sequencing). Additionally, this project will extend the type of epigenetic mechanisms to be studied beyond DNA methylation, specifically, microRNA expression. Hypotheses: Removal of mechanical loading or, oppositely, elevations of mechanical loading induce pro-bone resorbing or pro-bone forming changes in bone cell gene expression, respectively, via epigenetic mechanisms. Project 2, Title: Skeletal Sex Hormone Receptors and Simulated Spaceflight. Description and Aim: This work complements an upcoming space mission (Principal Investigator (PI) : Lane Christenson, former PI Joseph Tash, Kansas University of Medical Center-KUMC, co-I April Ronca) and will define the skeletal response of female mice to simulated long-duration spaceflight in the context of sex hormones and their receptor levels in leg bones. The biological assays used included gene expression or protein quantification in bone and blood. Additionally, we will investigate cage effects on estrous cyclicity. Finally, we aim to develop an artificial-intelligence algorithm for high-throughput quantification of cell imagery of the mouse estrous cycle. Hypotheses: Mechanical disuse reduces the levels of circulating sex hormones and their receptor levels in bone. These changes contribute to understanding skeletal deterioration during long-duration spaceflight. As part of this project, we proposed for and were awarded lumbar vertebrae from the Rodent Research 1 mission from the Life Sciences Data Archive (LSDA). We were able to quantify the cancellous bone changes from spaceflight, which will aid in future mission planning with Lane Christenson and Joe Tash. Project 3, Title: Investigating the effects of ionizing-radiation exposure on bone’s collagen mechanical properties. Description: Astronauts embarking on a deep-space mission to Mars will be exposed to high levels of ionizing radiation. This is of potential concern since ionizing radiation has been shown in other applications, such as radiotherapy treatment, to increase risk of fracture. While spaceflight-relevant radiation experiments so far have shown a quantitative degeneration of the tissue through loss of tissue mass, here we assessed the effects of such radiation on tissue quality. Hypothesis: We hypothesized that radiation-induced changes in collagen crosslinks would decrease the ability of mouse vertebrae to withstand cyclic mechanical loading. For this project, I co-mentored an NASA Space Technology Fellow and Doctoral Candidate from University of California Berkeley, Megan Pendleton. (Megan’s fellowship covers her stipend, but does not cover animal procurement and housing. Thus, I am covering the latter and some related expenses.) This is a collaboration with Tony Keaveny (UC Berkeley) and Jean Sibonga (Johnson Space Center-JSC). Project 4, Title: The role of Nuclear Cytoskeleton in the Osteocytic response to Simulated Weightlessness. Description and Aim: This project aims to investigate the role of nuclear-cytoskeletal elements (LINC, Linker of Nucleoskeleton and Cytoskeleton complexes) and related signaling cascades in the skeletal response to simulated weightlessness and radiation exposure. Description: This is a collaboration with Gunes Uzer, Boise State Univ. We investigated gene expression levels indicative of changes in this critical and emerging cytoskeletal complex. Project 5, Title: Characterizing the effects of chronic 2G centrifugation on the rat skeletal system. Description: This is a collaboration with Charles Fuller (UC Davis) and April Ronca (NASA Ames Research Center-ARC). Our long-term plan is to assess artificial gravity (AG) as a possible solution to mitigate these changes. In this study, one of our aims is to determine the skeletal acclimation to chronic centrifugation. Hypothesis: We hypothesize that a 2G hypergravity environment causes an anabolic response in growing male rats. Specifically, we predict chronic 2G to increase tissue mineral density, bone volume fraction of the cancellous tissue, and to increase overall bone strength. Systemically, we predict that bone formation markers (i.e., osteocalcin) are elevated and resorption markers (i.e., tartrate resistant acid phosphatase) are decreased or unchanged from controls. Projects conducted under this award addressed multiple Research Recommendations for Animal Biology and Developmental & Reproductive Biology stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the bone loss that astronauts experience while irradiated or weightless. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals, and may improve our fundamental understanding of skeletal regulation and the organism’s response to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Projects conducted under this award addressed multiple Research Recommendations for Animal Biology and Developmental & Reproductive Biology stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the bone loss that astronauts experience while irradiated or weightless. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals, and may improve our fundamental understanding of skeletal regulation and the organism’s response to spaceflight.

Task Progress:	This Presidential Early Career Award for Scientist and Engineers (PECASE) award consisted of five sub-projects, described earlier. Progress was made on each project, resulting in two peer-reviewed publications, manuscripts in preparation, and abstracts presented at national conferences. See Cumulative Bibliography section.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Jong A, Scott RT, Christenson LK, Ronca AE, Lindsey T, Alwood JS. "Deep Learning Classification of Estrous Stages in Mice." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Lowe MG, Kalotia SK, Tash JS, Christenson LK, Alwood JS, Ronca AE. "Female fertility during spaceflight: Ground-based analog studies of effects of cage type and calorie availability on estrous cyclicity in mice." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Delgadillo Talburt E, Woods K, Alwood JS, Unez G. "Does simulated spaceflight alter YAP/TAZ signaling in manner dependent on the LINC complex of the nuclear envelope?" Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Wu T, Pendleton MM, Bonnheim NB, Alwood JS, Keaveny TM. "The effect of in vivo ionizing radiation on the micromechanics of mouse vertebrae." To be presented at the 2019 Summer Biomechanics, Bioengineering, and Biotransport Conference, Seven Springs, PA, June 25 – 28, 2019. 2019 Summer Biomechanics, Bioengineering, and Biotransport Conference, Seven Springs, PA, June 25 – 28, 2019. , Jun-2019
Abstracts for Journals and Proceedings	Emerzian SR, Pendleton MM, O'Connell GD, Alwood JS, Keaveny TM. "Ionizing Radiation from Ex Vivo Sterilization Diminishes Fatigue but Not Static Murine Vertebral Body Mechanics." Presented at the 2019 Orthopedic Research Society, Austin, Texas, February 2-5, 2019. 2019 Orthopedic Research Society, Austin, Texas, February 2-5, 2019. , Feb-2019
Abstracts for Journals and Proceedings	Jong A, Scott RT, Christenson LK, Ronca AE, Lindsey T, Alwood JS. "Deep Learning Classification of Estrous Stages in Mice." To be presented at the Society for Study of Reproduction 52nd Annual Meeting, San Jose, CA, July 18-21, 2019. Society for Study of Reproduction 52nd Annual Meeting, San Jose, CA, July 18-21, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Lowe MG, Kalotia SK, Scott RT, Tash JS, Christenson LK, Alwood JS, Ronca AE. "Impact of Cage Type and Caloric Availability on Estrous Cyclicity in Mice." To be presented at the Society for Study of Reproduction 52nd Annual Meeting, San Jose, CA, July 18-21, 2019. Society for Study of Reproduction 52nd Annual Meeting, San Jose, CA, July 18-21, 2019. , Jul-2019
Articles in Peer-reviewed Journals	Pendleton MM, Emerzian SR, Liu J, Tang SY, O'Connell GD, Alwood JS, Keaveny TM. "Effects of ex vivo ionizing radiation on collagen structure and whole-bone mechanical properties of mouse vertebrae." Bone. 2019 Nov;128:115043. [2019 Aug 21 Epub ahead of print] https://doi.org/10.1016/j.bone.2019.115043 ; PubMed PMID: 31445224 , Nov-2019
Articles in Peer-reviewed Journals	Axpe E, Chan D, Abegaz MF, Schreurs AS, Alwood JS, Globus RK, Appel EA. "A human mission to Mars: Predicting the bone mineral density loss of astronauts." PLoS ONE. 2020 Jan 22;15(1):e0226434. https://doi.org/10.1371/journal.pone.0226434 ; PubMed PMID: 31967993 , Jan-2020
Articles in Peer-reviewed Journals	Pendleton MM, Emerzian SR, Sadoughi S, Li A, Liu JW, Tang SY, O'Connell GD, Sibonga JD, Alwood JS, Keaveny TM. "Relations between bone quantity, microarchitecture, and collagen cross-links on mechanics following in vivo irradiation in mice." JBMR Plus. 2021 Nov;5(11):e10545. https://doi.org/10.1002/jbm4.10545 ; PMID: 34761148; PMCID: PMC8567491 , Nov-2021
Articles in Peer-reviewed Journals	Pendleton MM, Sadoughi S, Li A, O'Connell GD, Alwood JS, Keaveny TM. "High-precision method for cyclic loading of small-animal vertebrae to assess bone quality." Bone Reports. 2018 Oct 27;9:165-72. https://doi.org/10.1016/j.bonr.2018.10.002 ; PubMed PMID: 30417036; PubMed Central PMCID: PMC6222041 , Oct-2018
Articles in Peer-reviewed Journals	Touchstone H, Bryd R, Loisate S, Thompson M, Kim S, Puranam K, Senthilnathan AN, Pu X, Beard R, Rubin J, Alwood J, Oxford JT, Uzer G. "Recovery of stem cell proliferation by low intensity vibration under simulated microgravity requires LINC complex." npj Microgravity. 2019 May 15;5(1):11. 2019 May 15;5:11. eCollection 2019. https://doi.org/10.1038/s41526-019-0072-5 ; PubMed PMID: 31123701; PubMed Central PMCID: PMC6520402 [NOTE originally reported as "Accepted for publication in npj Microgravity and in revision as of March 4, 2019.'] , May-2019